Labrorator 2 Algoritmi fundamentali
MEMORAREA GRAFURILOR. PARCURGERI. APLICAȚII

Implementați algoritmi eficienți pentru rezolvarea următoarelor probleme. Se vor adăuga comentarii în cod cu ideea algoritmului și complexitatea sa.

1. **Graf bipartit.** https://cses.fi/problemset/task/1668

Building Teams

Time limit: 1.00 s **Memory limit:** 512 MB

There are n pupils in Uolevi's class, and m friendships between them. Your task is to divide the pupils into two teams in such a way that no two pupils in a team are friends. You can freely choose the sizes of the teams.

Input

The first input line has two integers n and m: the number of pupils and friendships. The pupils are numbered $1, 2, \ldots, n$.

Then, there are m lines describing the friendships. Each line has two integers a and b: pupils a and b are friends.

Every friendship is between two different pupils. You can assume that there is at most one friendship between any two pupils.

Output

Print an example of how to build the teams. For each pupil, print "1" or "2" depending on to which team the pupil will be assigned. You can print any valid team.

If there are no solutions, print "IMPOSSIBLE".

Constraints

- $1 \le n \le 10^5$
- $1 \le m \le 2 \cdot 10^5$
- $1 \le a, b \le n$

Example

Input:

```
5 3
1 2
1 3
4 5
```

Output:

```
1 2 2 1 2
```

2. a) **Sortare topologică.** https://cses.fi/problemset/task/1679

Course Schedule

Time limit: 1.00 s **Memory limit:** 512 MB

You have to complete n courses. There are m requirements of the form "course a has to be completed before course b". Your task is to find an order in which you can complete the courses.

Input

The first input line has two integers n and m: the number of courses and requirements. The courses are numbered $1, 2, \ldots, n$.

After this, there are m lines describing the requirements. Each line has two integers a and b: course a has to be completed before course b.

Output

Print an order in which you can complete the courses. You can print any valid order that includes all the courses.

If there are no solutions, print "IMPOSSIBLE".

Constraints

- $1 \leq n \leq 10^5$
- $1 \leq m \leq 2 \cdot 10^5$
- $1 \leq a, b \leq n$

Example

Input:

```
5 3
1 2
3 1
4 5
```

Output:

```
3 4 1 5 2
```

b) Modificați programul de la a) astfel încât în cazul în care nu pot fi urmate toate cursurile (deci se afișează IMPOSSIBLE), să afișeze și o listă de cursuri [c_1, c_2, …, c_k, c_1] care depind circular unele de altele (orice curs c_i trebuie urmat înaintea cursului c_{i-1}, iar c_k înaintea cursului c_1).

3. **Fill** https://cses.fi/problemset/task/1192

Counting Rooms

Time limit: 1.00 s **Memory limit:** 512 MB

You are given a map of a building, and your task is to count the number of its rooms. The size of the map is $n \times m$ squares, and each square is either floor or wall. You can walk left, right, up, and down through the floor squares.

Input

The first input line has two integers n and m: the height and width of the map.

Then there are n lines of m characters describing the map. Each character is either . (floor) or # (wall).

Output

Print one integer: the number of rooms.

Constraints

- $1 \leq n, m \leq 1000$

Example

Input:

```
5 8
########
#..#...#
####.#.#
#..#...#
########
```

Output:

```
3
```

4. **Componente tare conexe.** https://cses.fi/problemset/task/1683

Planets and Kingdoms

Time limit: 1.00 s **Memory limit:** 512 MB

A game has n planets, connected by m teleporters. Two planets a and b belong to the same kingdom exactly when there is a route both from a to b and from b to a. Your task is to determine for each planet its kingdom.

Input

The first input line has two integers n and m: the number of planets and teleporters. The planets are numbered $1, 2, \ldots, n$.

After this, there are m lines describing the teleporters. Each line has two integers a and b: you can travel from planet a to planet b through a teleporter.

Output

First print an integer k: the number of kingdoms. After this, print for each planet a kingdom label between 1 and k. You can print any valid solution.

Constraints

- $1 \leq n \leq 10^5$
- $1 \leq m \leq 2 \cdot 10^5$
- $1 \leq a, b \leq n$

Example

Input:

```
5 6
1 2
2 3
3 1
3 4
4 5
5 4
```

Output:

```
2
1 1 1 2 2
```

5. **Puncte critice (de articulație).** https://www.spoj.com/problems/SUBMERGE/

SUBMERGE - Submerging Islands

Vice City is built over a group of islands, with bridges connecting them. As anyone in Vice City knows, the biggest fear of vice-citiers is that some day the islands will submerge. The big problem with this is that once the islands submerge, some of the other islands could get disconnected. You have been hired by the mayor of Vice city to tell him how many islands, when submerged, will disconnect parts of Vice City. You should know that initially all the islands of the city are connected.

Input

The input will consist of a series of test cases. Each test case will start with the number N ($1 \le N \le 10^4$) of islands, and the number M of bridges ($1 \le M \le 10^5$). Following there will be M lines each describing a bridge. Each of these M lines will contain two integers Ui, Vi ($1 \le Ui, Vi \le N$), indicating that there is a bridge connecting islands Ui and Vi. The input ends with a case where N = M = 0.

Output

For each case on the input you must print a line indicating the number of islands that, when submerged, will disconnect parts of the city.

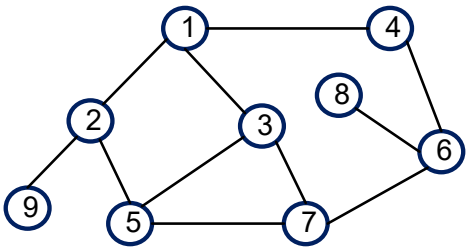
Example

Input:

```
3 3
1 2
2 3
1 3
6 8
1 3
6 1
6 3
4 1
6 4
5 2
3 2
3 5
0 0
```

Output:

```
0
1
```

6. **Distanțe și drumuri minime**. Se dă o rețea neorientată cu n noduri și o listă de noduri reprezentând puncte de control pentru rețea în fișierul ***graf.in***, în formatul precizat la începutul laboratorului; în plus, pe ultima linie din fișier se află punctele de control separate prin spațiu. Să se determine pentru fiecare nod din rețea distanța până la cel mai apropiat punct de control de acesta. În fișierul ***graf.out*** se vor afișa pentru fiecare nod de la 1 la n aceste distanțe separate prin spațiu.

graf.in - graful alaturat; pe ultima linie: punctele de control	**graf.out**
9 11 1 2 1 3 1 4 2 5 2 9 3 5 3 7 5 7 6 7 6 8 4 6 **8 9**	2 1 3 2 2 1 2 0 0 **(Explicații:** **Punctele de control sunt 8 si 9** **- cel mai apropiat punct de control de nodul 1 este 9, aflat la distanță 2** **- cel mai apropiat punct de control de nodul 2 este 9, aflat la distanță 1** etc)